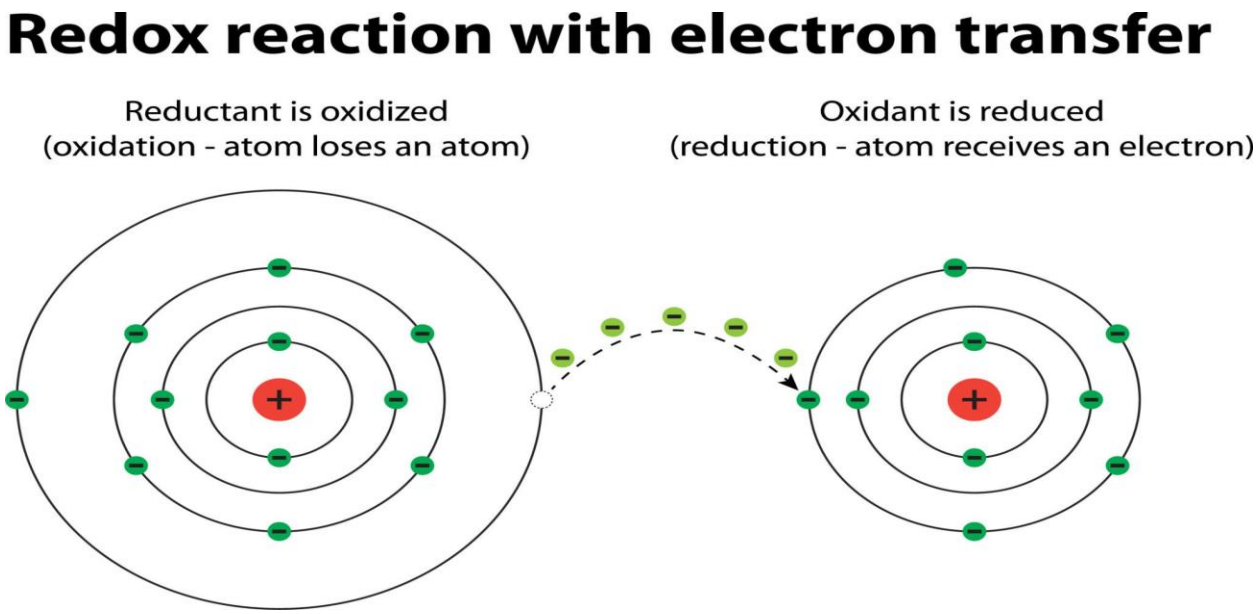
REDOX REACTIONS

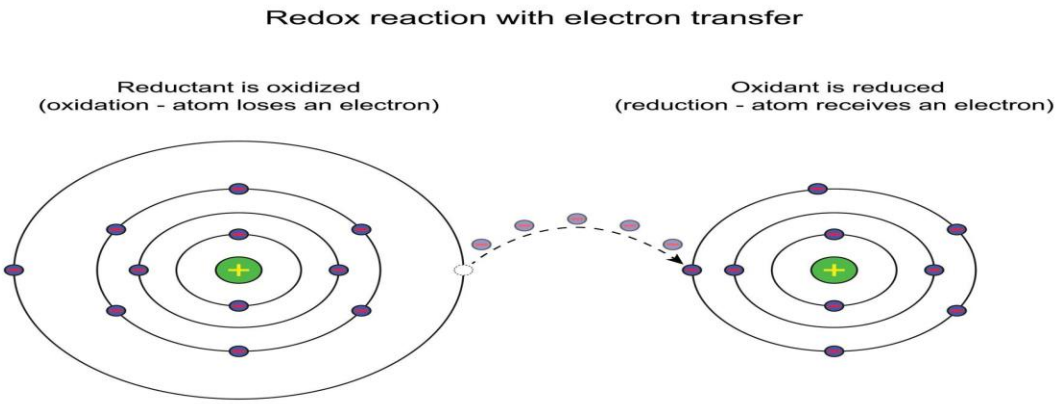
Introduction

Redox reactions are chemical reactions in which oxidation and reduction occur simultaneously. These reactions involve the transfer of electrons between species and are essential in processes such as respiration, corrosion, and electrochemical cells. Understanding redox reactions is important for NEET as they form the basis of many chemical processes.

Concept of Oxidation and Reduction

Oxidation is defined as the loss of electrons, while reduction is the gain of electrons. In any redox reaction, one species is oxidized and another is reduced.





These processes always occur together.

Oxidizing and Reducing Agents

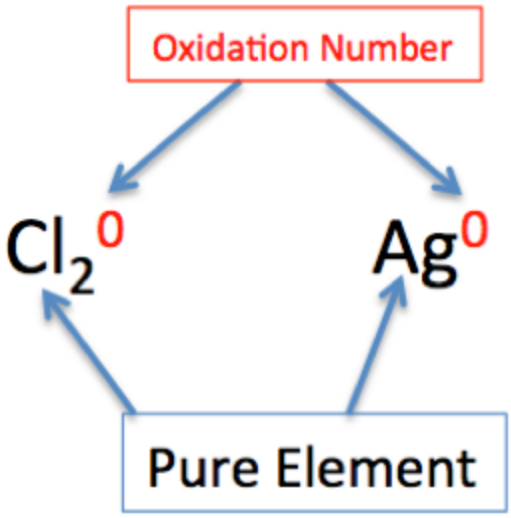
An oxidizing agent is a substance that causes oxidation by accepting electrons, while a reducing agent causes reduction by donating electrons.

The strength of these agents determines the direction of a redox reaction.

Oxidation Number

Oxidation number represents the charge assigned to an atom in a compound. It helps in identifying oxidation and reduction processes.

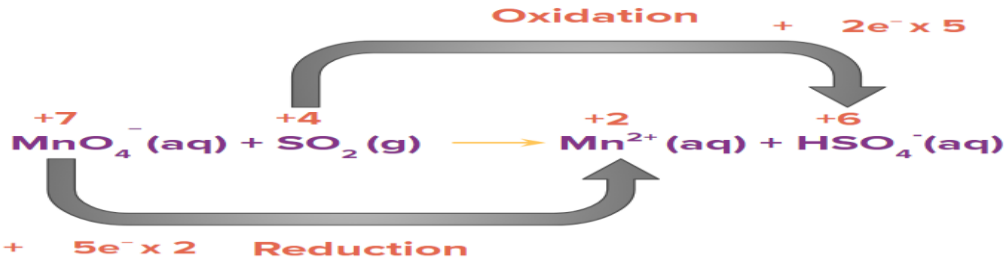
Pure Element	Oxidation Number
Ag	Zero
He	
Cu	
Cl_2	
O_2	
S_8	

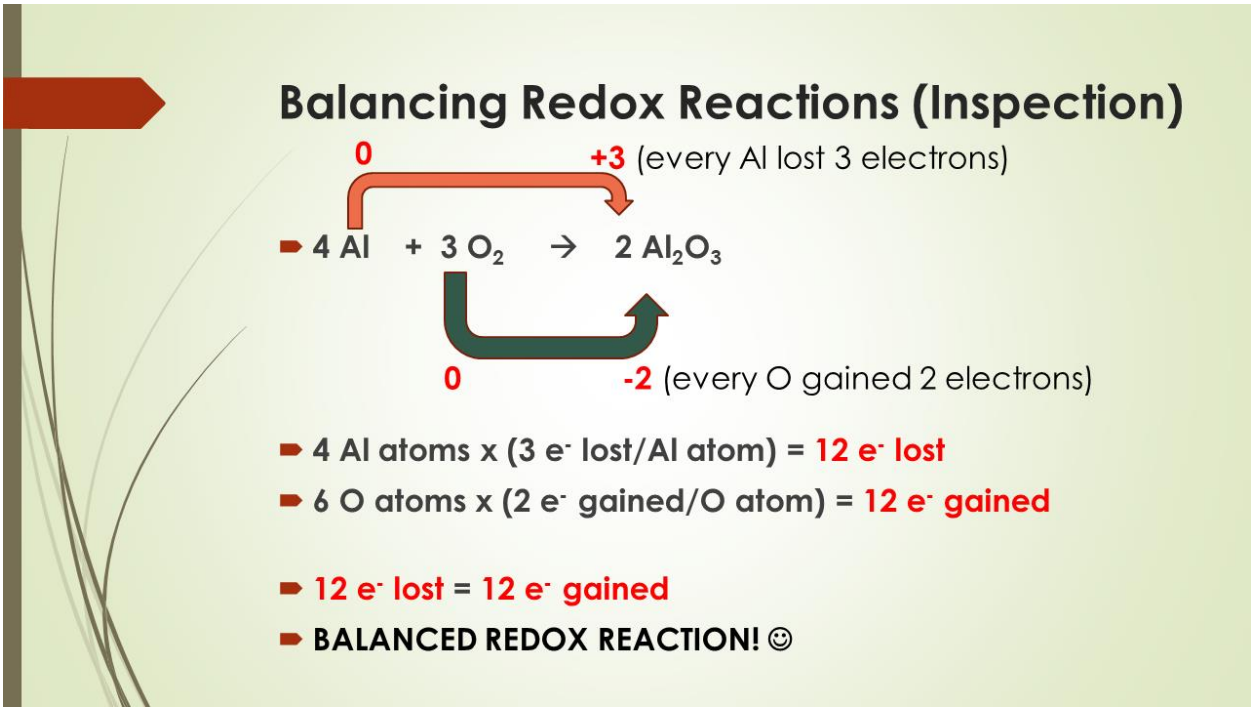


Changes in oxidation number indicate redox reactions.

Balancing Redox Reactions

Redox reactions are balanced using methods such as the oxidation number method and the ion-electron method. These methods ensure conservation of mass and charge.





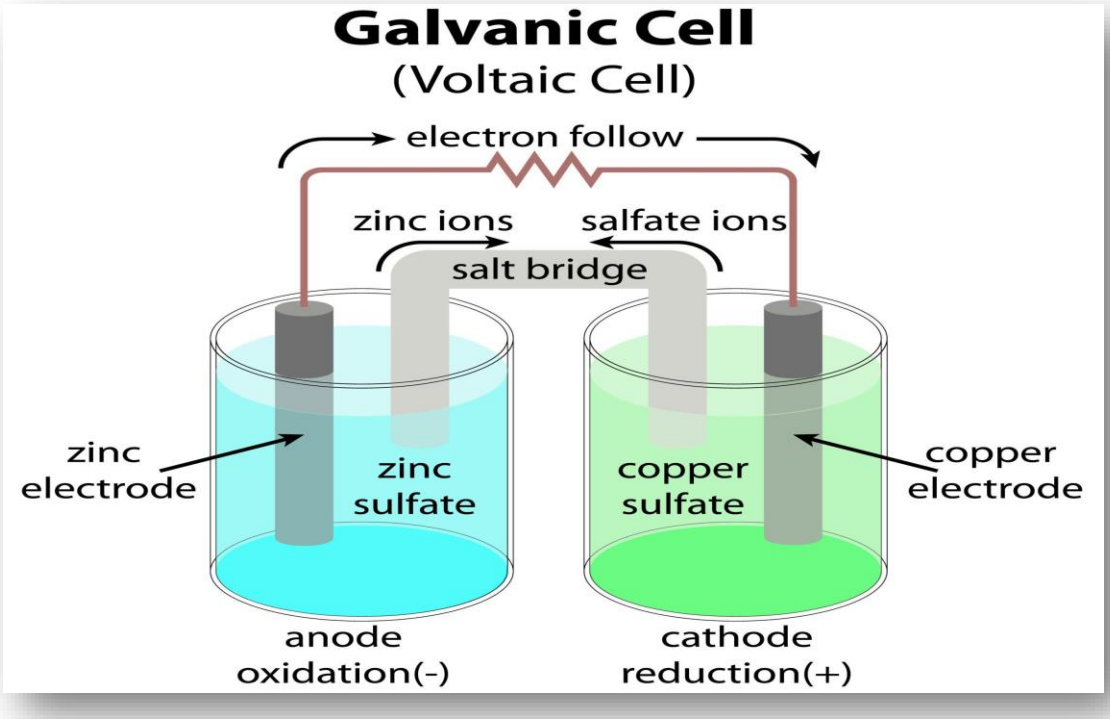
Balancing is essential for accurate chemical equations.

Types of Redox Reactions

Redox reactions can be classified into combination, decomposition, displacement, and disproportionation reactions based on the nature of the process.

Electrochemical Cells

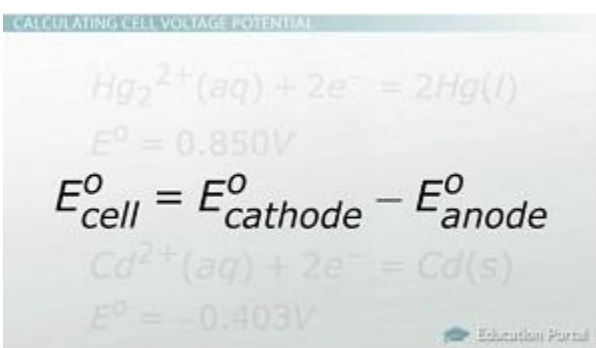
Electrochemical cells convert chemical energy into electrical energy through redox reactions. They consist of two electrodes and an electrolyte.

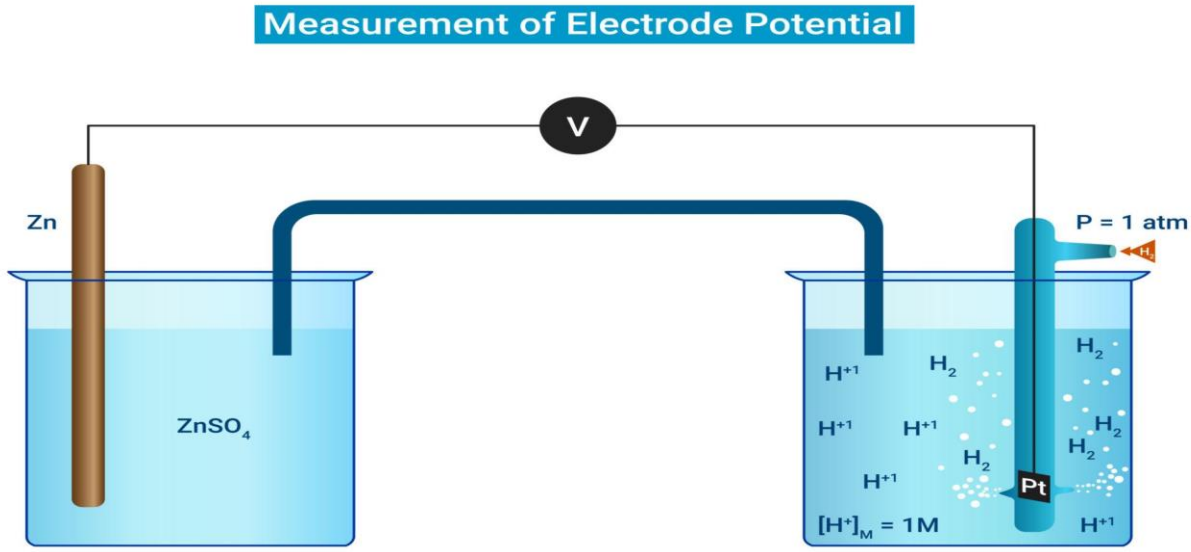


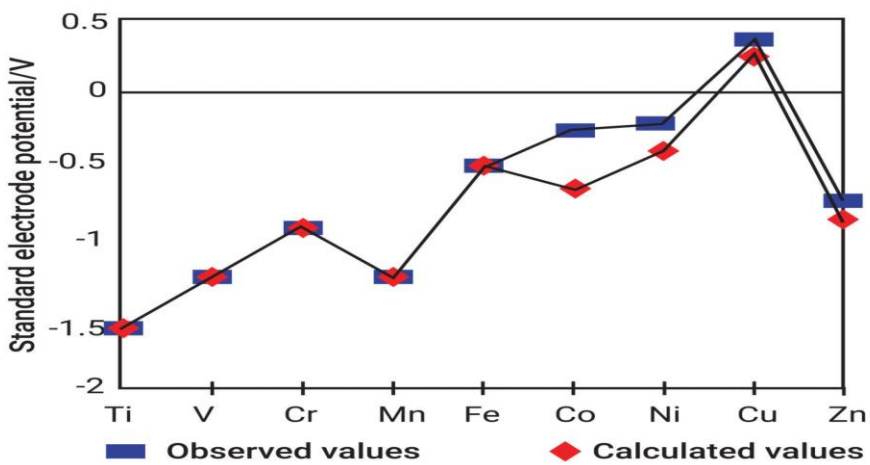
hese cells are widely used in batteries and energy systems.

Standard Electrode Potential

Standard electrode potential measures the tendency of a species to be reduced. It is used to determine the feasibility of redox reactions.







Higher potential indicates stronger oxidizing ability.

Applications of Redox Reactions

Redox reactions are involved in various processes such as corrosion, respiration, photosynthesis, and industrial reactions. They play a crucial role in energy production and environmental processes.

Conclusion

Redox reactions involve transfer of electrons and are fundamental to many chemical and biological processes. Understanding oxidation numbers, balancing methods, and electrochemical cells is essential for NEET preparation. This chapter is highly scoring and requires strong conceptual clarity.